

ID ()

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
5	3	4	1	2	2	0	0	0	1	4	4	1	3	1	$\frac{\pi}{4}$	$\frac{\pi}{2}$

18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34
$\frac{\pi}{2}$	π	4π	0	2	3	6	5	1	0	2	4	-2	-6	0	2	F

[1-3] Express each of the following complex numbers in polar form, $re^{j\theta}$.
Please fill in the answer table.

- [1] $-1 - j$
- (1) $r = 1, \theta = \pi/4$ (2) $r = \sqrt{2}, \theta = \pi/2$ (3) $r = 1, \theta = -\pi/2$
(4) $r = \sqrt{2}, \theta = \pi/4$ (5) $r = \sqrt{2}, \theta = -3\pi/4$

- [2] $(1 - j)^2$
- (1) $r = \sqrt{2}, \theta = 3\pi/4$ (2) $r = \sqrt{2}, \theta = \pi/2$ (3) $r = 2, \theta = -\pi/2$
(4) $r = 1, \theta = -3\pi/4$ (5) $r = 1, \theta = 3\pi/4$

- [3] $j(1 - j)$
- (1) $r = 1, \theta = -\pi$ (2) $r = 2, \theta = 0$ (3) $r = \sqrt{2}, \theta = -\pi$
(4) $r = \sqrt{2}, \theta = \pi/4$ (5) $r = \sqrt{2}, \theta = \pi/2$

[4-6] Express each of the following complex numbers in rectangular form.
Please fill in the answer table.

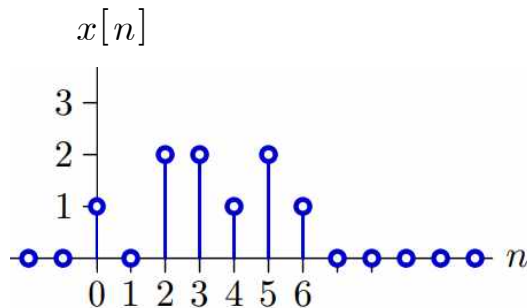
- [4] $\sqrt{2} e^{j\pi/4}$
- (1) $1+j$ (2) $-1+j$ (3) $-1-j$ (4) $1-j$ (5) $\sqrt{2}+j$

- [5] $\{\sqrt{2} e^{j\pi/4}\} \cdot \{e^{-j(7\pi/2)}\}$
- (1) $1+j$ (2) $-1+j$ (3) $\sqrt{2}+j$ (4) $-1+j\sqrt{2}$ (5) $\sqrt{2}-j\sqrt{2}$

- [6] $e^{j\pi/4} + e^{-j\pi/4}$
- (1) 1 (2) $\sqrt{2}$ (3) $-1+j$ (4) $-j\sqrt{2}$ (5) $j\sqrt{2}$

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[7-15] A discrete-time signal $x[n]$ is shown below,



When $y[n] = (x[5-n] + x[n]) u[n-1]$,

find the values of $y[n]$ and fill in the answer table.

[7] $y[-2]$

[8] $y[-1]$

[9] $y[0]$

[10] $y[1]$

[11] $y[2]$

[12] $y[3]$

[13] $y[4]$

[14] $y[5]$

[15] $y[6]$

[16-20] Consider a continuous-time signal $x(t) = \cos(\omega_a t)$.

$x(t)$ is converted to the discrete-time signal $x[n] = \cos(\omega_d n)$ with the sampling frequency f_s . Find the digital frequency ω_d [rad] of the sampled signal in each case, and fill in the answer table.

[16] $\omega_a = 2000 \pi$ [rad/sec] , $f_s = 8$ [kHz]

[17] $\omega_a = 2000 \pi$ [rad/sec] , $f_s = 4$ [kHz]

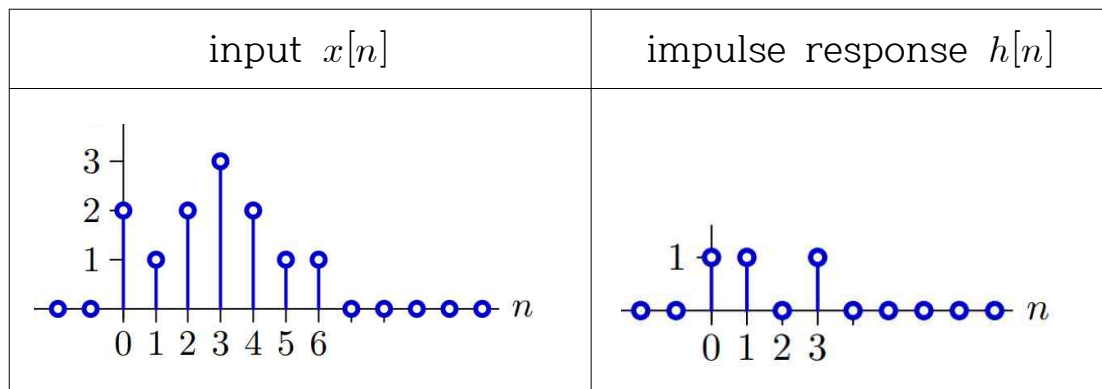
[18] $\omega_a = 4000 \pi$ [rad/sec] , $f_s = 8$ [kHz]

[19] $\omega_a = 8000 \pi$ [rad/sec] , $f_s = 8$ [kHz]

[20] $\omega_a = 4,000 \pi$ [rad/sec] , $f_s = 1$ [kHz]

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[21-27] Consider an LTI system with impulse response $h[n]$ and input $x[n]$. Find the output $y[n]$ and fill in the answer table.



[21] $y[-2]$

[22] $y[0]$

[23] $y[2]$

[24] $y[4]$

[25] $y[6]$

[26] $y[8]$

[27] $y[10]$

[28-34] When $x[n] = \delta[n] + 2\delta[n-1] - \delta[n-3]$ is the input to a system and $h[n] = 2\delta[n+1] - 2\delta[n-1]$ is the impulse response of the system. Compute output $y[n] = x[n] * h[n]$ and fill in the answer table.

[28] $y[-1]$

[29] $y[0]$

[30] $y[1]$

[31] $y[2]$

[32] $y[3]$

[33] $y[4]$

[34] This system is a causal system. True[T] or False[F]?